

Power BI – Module End Evaluation

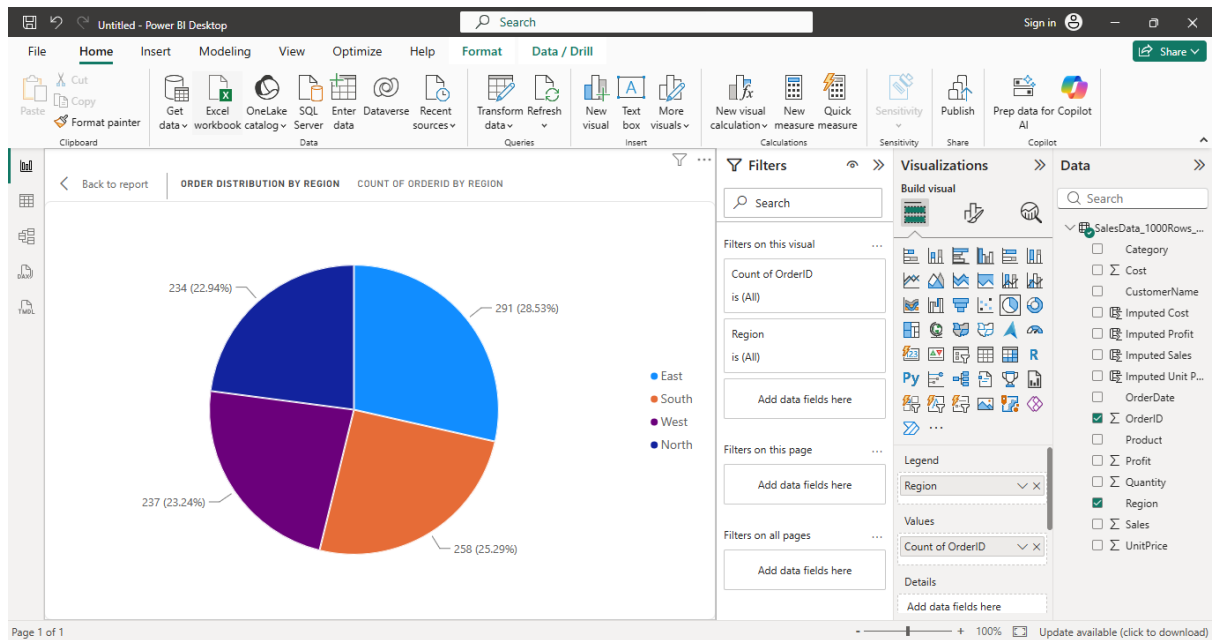
Dataset: SalesData_1000Rows_WithIssues_copy.csv

Q1: Data Cleaning and Loading

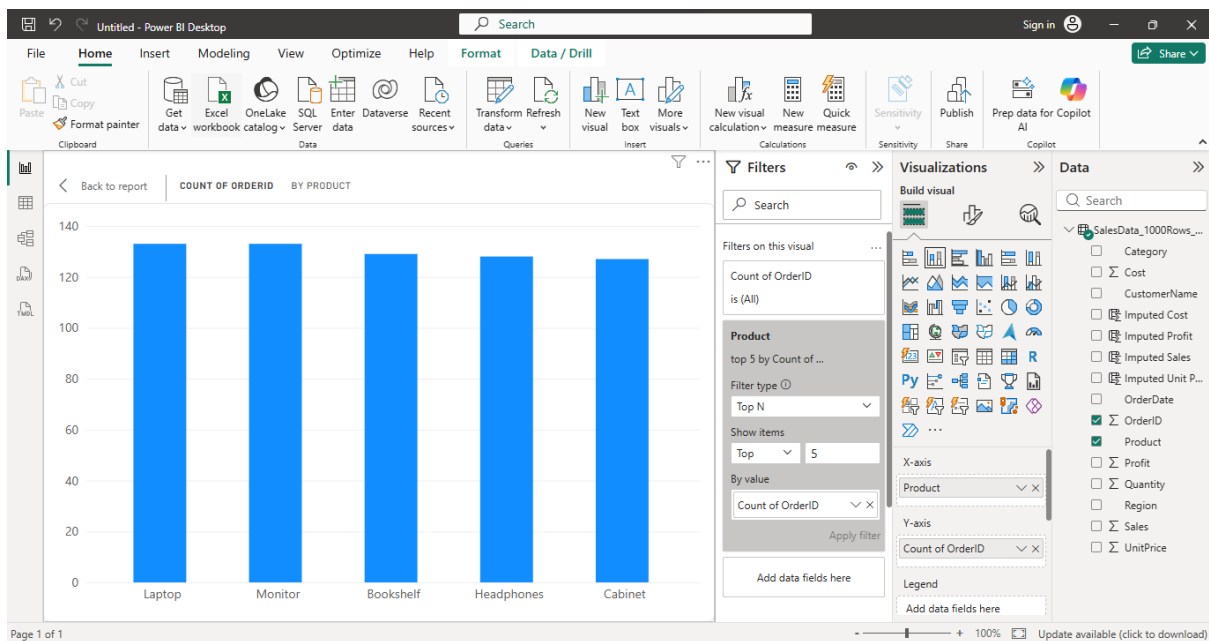
- a) Imported the given file using 'Get Data' in 'Home' Menu
 - b) Opened the power query by using 'Transform Data'
 - i) Handling missing values in columns.
 - Column Profile and Column quality option in view mwnu is used to identify the distribution of empty cells in each column
 - Null names were replaced with 'Unknown' using 'replace values' in 'Transform' Menu
 - Null values in date are avoided so that it won't cause trouble for us in data analysis and does not affect the date format.
 - As there is no single formula to find the mode to replace blank texts in the 'region' column, text mode is manually identified with the help of filters and replaced using replace values.
 - In product column, blank cells is replaced with unknown using replace values.
 - Unit price, sales, Cost, Profit columns were imputed by using calculated columns in DAX and using 'Coalesce' function.
 - ii) Removing duplicate rows.
 - Duplicate rows are removed using Remove rows – remove duplicate rows in power query.
 - iii) Standardizing the OrderDate column to a consistent date format.
 - Order date format is standardized using change type with locale option using format icon on the column title.
 - c) Load the cleaned dataset into the Power BI model.
 - Use close and apply in power query to bring the changes to powerBI.
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Q2: Visualizations and Insights

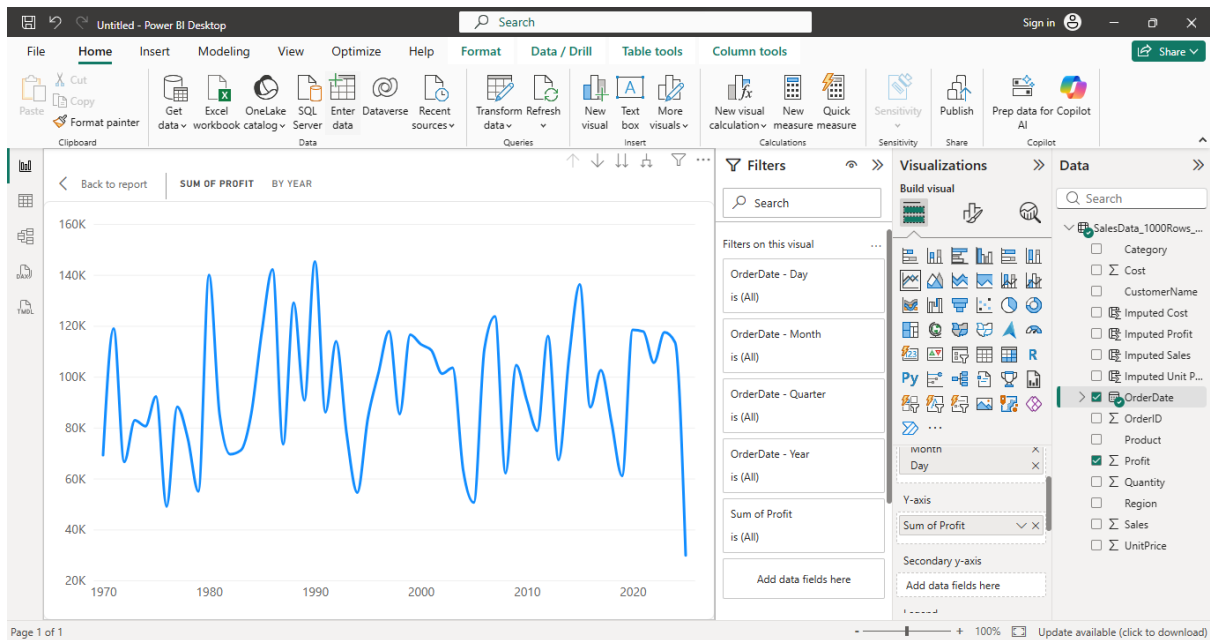
- a) In the report view a pie chart is inserted, in Legend region is dragged and in values the order ID is added to get the pie chart.



b) Insert a column chart, on x axis, add product, on y axis add count of orderID, on the filter Pane select TopN filter type and select based on count of order ID.



c) Insert Line chart, on X axis put the date, and on Y axis put Profit.



Q3: DAX Calculations (5 Marks)

Apply DAX to create calculated elements and gain deeper insights.

- a) use New table option in home menu and use 'Filter' function in formula bar to filter all the details regarding the east region.

OrderID	OrderDate	CustomerName	Region	Product	Category	Quantity	UnitPrice	Sales	Cost	Profit	Imputed Unit Price	Imputed Profit
1586	3/14/2025 12:00:00 AM	Richard Aguirre	East	Bookshelf	Furniture	5	53758	268790	258904	9886	53758	53758
1895	12/23/1985 12:00:00 AM	Nathan Maldonado	East	Cabinet	Furniture	5	47731	238655	231266	7389	47731	47731
1599	5/11/1985 12:00:00 AM	Jordan Henderson	East	Cabinet	Furniture	5	58403	292015	290925	1090	58403	58403
1064	5/3/1992 12:00:00 AM	Unknown	East	Laptop	Electronics	5	56571	282855	274397	8458	56571	56571
1754	12/4/2003 12:00:00 AM	Ryan Munoz	East	Cabinet	Furniture	5	46776	233880	232440	1440	46776	46776
1045	10/31/1980 12:00:00 AM	Gerald Hensley	East	Smartphone	Electronics	5	50965	254825	253344	1481	50965	50965
1536	4/23/2019 12:00:00 AM	Gabrielle Davis	East	Monitor	Electronics	5	25648	128240	126825	1415	25648	25648
1811	2/16/1993 12:00:00 AM	Stephen McKee	East	Laptop	Electronics	5	27401	137005	128677	8328	27401	27401
1967	11/19/2002 12:00:00 AM	Victoria Garcia	East	Desk Chair	Furniture	5	44483	222415	220439	1976	44483	44483
1823	2/11/1980 12:00:00 AM	Fred Smith	East	Cabinet	Furniture	5	43999	219995	216282	3713	43999	43999
1833	12/26/2000 12:00:00 AM	Michael Evans	East	Cabinet	Furniture	5	58274	291370	285777	5593	58274	58274
1179	11/20/2024 12:00:00 AM	Danny Morgan	East	Headphones	Electronics	5	54115	270575	266872	3703	54115	54115
1449	12/7/1991 12:00:00 AM	Jordan Henderson	East	Headphones	Electronics	5	39129	195645	189014	6631	39129	39129
1944	4/10/2001 12:00:00 AM	Mike Allen	East	Monitor	Electronics	5	21274	106370	98674	7696	21274	21274
1635	3/22/1972 12:00:00 AM	Anthony Humphrey	East	Headphones	Electronics	5	51416	257080	253578	3502	51416	51416
1240	8/22/1987 12:00:00 AM	Brent Jordan	East	Bookshelf	Furniture	5	15504	77520	74559	2961	15504	15504
1751	8/15/1994 12:00:00 AM	Elizabeth Fowler	East	Monitor	Electronics	5	26874	134370	129301	5069	26874	26874
1953	10/1/1980 12:00:00 AM	Danny Morgan	East	Laptop	Electronics	5	26317	131585	122396	9189	26317	26317
1724	6/11/1992 12:00:00 AM	Aimee Montoya	East	Desk Chair	Furniture	5	7610	38050	32844	5206	7610	7610
1595	4/6/2011 12:00:00 AM	Christina Walters	East	Office Desk	Furniture	5	57017	285085	283974	1111	57017	57017
1312	4/24/2024 12:00:00 AM	Jessica Holmes	East	Laptop	Electronics	5	41264	206320	204943	1377	41264	41264
1465	5/17/1982 12:00:00 AM	John Pierce	East	Laptop	Electronics	5	2686	13430	11615	1815	2686	2686

- b) Use new column to add a new column, and on the formula bar add the equation.

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File Home Help Table tools Column tools

Name Profit cost difference Format Whole number Summarization Sum Data category Uncategorized

Data type Whole number \$ % 0

Sort by column Data groups Manage relationships New column

Structure Formatting Properties Calculations

1 Profit cost difference = 'East Region Orders'[Cost] - 'East Region Orders'[Profit]

Category	Quantity	UnitPrice	Sales	Cost	Profit	Imputed Unit Price	Imputed Sales	Imputed Cost	Imputed Profit	Profit cost difference
Furniture	5	53758	268790	258904	9886	53758	268790	258904	9886	249018
Furniture	5	47731	238655	231266	7389	47731	238655	231266	7389	223877
Furniture	5	58403	292015	290925	1090	58403	292015	290925	1090	289835
Electronics	5	56571	282855	274397	8458	56571	282855	274397	8458	265939
Furniture	5	46776	233880	232440	1440	46776	233880	232440	1440	231000
Electronics	5	50965	254825	253344	1481	50965	254825	253344	1481	251863
Electronics	5	25648	128240	126825	1415	25648	128240	126825	1415	125410
Electronics	5	27401	137005	128677	8328	27401	137005	128677	8328	120349
Furniture	5	44483	222415	220439	1976	44483	222415	220439	1976	218463
Furniture	5	43999	219995	216282	3713	43999	219995	216282	3713	212569
Furniture	5	58274	291370	285777	5593	58274	291370	285777	5593	280184
Electronics	5	54115	270575	266872	3703	54115	270575	266872	3703	263169
Electronics	5	39129	195645	189014	6631	39129	195645	189014	6631	182383
Electronics	5	21274	106370	98674	7696	21274	106370	98674	7696	90978
Electronics	5	51416	257080	253578	3502	51416	257080	253578	3502	250076
Furniture	5	15504	77520	74559	2961	15504	77520	74559	2961	71598
Electronics	5	26874	134370	129301	5069	26874	134370	129301	5069	124232
Electronics	5	26317	131585	122396	9189	26317	131585	122396	9189	113207
Furniture	5	7610	38050	32844	5206	7610	38050	32844	5206	27638
Furniture	5	57017	285085	283974	1111	57017	285085	283974	1111	282863
Electronics	5	41264	206320	204943	1377	41264	206320	204943	1377	203566
Electronics	5	2686	13430	11615	1815	2686	13430	11615	1815	9800

Table: East Region Orders (291 rows) Column: Profit cost difference (287 distinct values)

Update available (click to download)

c) Use New Measure option in 'home' menu and type the formula using sum function.

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Search

File Home Help Table tools Measure tools

Name Total Profit Format General Data category Uncategorized

Home table East Region Orders \$ % Auto

New measure Quick measure measure

Structure Formatting Properties Calculations

1 Total Profit = SUM('East Region Orders'[Imputed Profit])

OrderID	OrderDate	CustomerName	Region	Product	Category	Quantity	UnitPrice	Sales	Cost	Profit	Imputed Unit Price	Imputed Sales	Imputed Cost	Imputed Profit
1586	3/14/2025 12:00:00 AM	Richard Aguirre	East	Bookshelf	Furniture	5	53758	268790	258904	9886	53758	268790	258904	9886
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1754	12/4/2003 12:00:00 AM	Ryan Munoz	East	Cabinet	Furniture	5	46776	233880	232440	1440	46776	233880	232440	1440
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1967	11/19/2002 12:00:00 AM	Victoria Garcia	East	Desk Chair	Furniture	5	44483	222415	220439	1976	44483	222415	220439	1976
1823	2/11/1980 12:00:00 AM	Fred Smith	East	Cabinet	Furniture	5	43999	219995	216282	3713	43999	219995	216282	3713
1833	12/26/2000 12:00:00 AM	Michael Evans	East	Cabinet	Furniture	5	58274	291370	285777	5593	58274	291370	285777	5593
1179	11/20/2024 12:00:00 AM	Danny Morgan	East	Headphones	Electronics	5	54115	270575	266872	3703	54115	270575	266872	3703
1449	12/7/1991 12:00:00 AM	Jordan Henderson	East	Headphones	Electronics	5	39129	195645	189014	6631	39129	195645	189014	6631
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1312	4/24/2024 12:00:00 AM	Jessica Holmes	East	Laptop	Electronics	5	41264	206320	204943	1377	41264	206320	204943	1377
1465	5/17/1982 12:00:00 AM	John Pierce	East	Laptop	Electronics	5	2686	13430	11615	1815	2686	13430	11615	1815

Table: East Region Orders (291 rows) Column: Total Profit (0 distinct values)

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